

Membrane Air Separation

Final Lab Report

Unit Operations Lab #3 

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Group 2 of Tuesday AM section

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1. Executive Summary

The objective of this experiment is to calculate the overall transport coefficients by observing the membrane separation process and analyzing it by evaluating its effect on the composition of air at the outlet of the system. Membrane separation is a pressure driven process and it is based on the relative rates of mass transfer that requires low energy consumption. The experiment is performed in parallel and series configuration. From the results for both configurations, it was concluded that the separation of air is consistent with literature. The experimental data shows that nitrogen recovery in series and parallel have constancy and for that of oxygen the parallel and series flow rates deviate slightly, which is shown in Figures 5 and 6. When comparing the data in our results, the coefficients and recoveries were higher when run in series. For example, Trial 2 at 25 psig for oxygen had a recovery of 0.870636 in parallel and 0.91363 in series.

In conclusion, the STAGECUT is higher for smaller flowrates, but the ratio of the oxygen coefficients to the nitrogen coefficients is lower for smaller flowrates; these trends appear in both for parallel and series. This means that at lower temperatures more of the feed stream passes through the membrane, but there is less separation of the oxygen from the nitrogen. Furthermore, both membrane separation configurations are effective, but series is the most optimal when looking at cost and capital expense for real world application.

2. Introduction

The purpose of this experiment is to calculate the overall transport coefficients. A membrane separation using a nonporous semi-permeable membrane is observed and the effect it has on the composition of air is analyzed. The apparatus consists of two Prism™ separated modules (columns) mounted on a horizontal thick acrylic sheet. The columns are operated in series or parallel. The modules have a diameter of one inch and a length of three feet [1].

Some industrial applications for membrane air separation is desalination to clean and purify wastewater, food industry, pharmaceutical and even biotechnological. The food industry implements membrane separation to increase the performance of starch and sweetener products and to clarify unprocessed juice in sugars. The pharmaceutical industry uses this method for fermentation primarily for antibiotics and producing enzymes [2].

3. Theory

Gas permeation is a method using nonporous membranes to fractionate gas mixtures of permeate and non-permeate streams. The membrane separation is caused by a pressure difference across the membrane [3]. The governing principles of the lab is based on Fick's Law which relates to diffusion and Henry's Law which relates to solubility.

Fick's Law states that the diffusion is proportional to the concentration and inverse to the thickness of the membrane.

$$J_A = D_A \left(\frac{C_{a1} - C_{a2}}{Z} \right)$$

Where

J_A is the flux of component A

D_A is the diffusivity of component A

Z is the thickness of the membrane

C_{a1} is the concentration of component A inside the membrane wall feed side

C_{a2} is the concentration of component A outside the membrane wall permeate side

Total Flux can be calculated by:

$$J_A = \frac{V_{Ap}\rho}{nA}$$

V_{Ap} is the volumetric flowrate of species A in permeate

ρ is the density of the permeate

A is the membrane area (2.7m² per module)

n is the number of modules used

Henry's Law states that the partial pressure of a gas is proportional to the mole fraction.

$$C_A = p_A S_A$$

Where

C_A is the concentration of component A inside membrane wall

S_A is the solubility constant for component A in the membrane

p_A is the partial pressure of component A in the gas phase

When analyzing the separation of the system, an experimental separation factor is calculated using mole fractions given by oxygen sensors. The separation factor is based on retentate composition and feed composition [1]. The separation factor based on the retentate composition is the ratio of the permeate composition divided by the ratio of the retentate composition, while the feed-based separation factor is the ratio of the permeate composition divided by the feed composition.

4. Results

The overall transport coefficients were derived from the transfer rates and partial pressures. As shown in **Appendix I, Table 1 and Table 2**, the overall transport coefficients for Oxygen remain generally consistent at different pressures for both series and parallel. The overall transport coefficients for Nitrogen seem to increase slowly as pressure increases for series and for parallel. No significant lack of constancy can be demonstrated for the transport coefficients of Oxygen and Nitrogen.

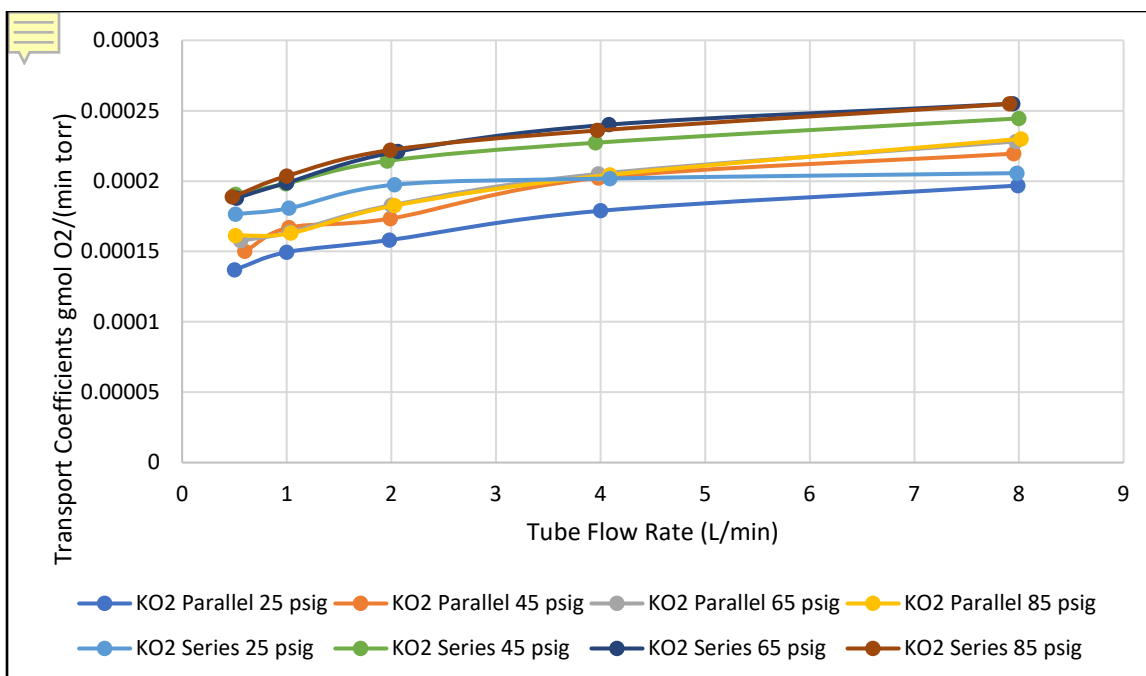


Figure 1: Oxygen transport coefficients (parallel and series) VS tube flow rate

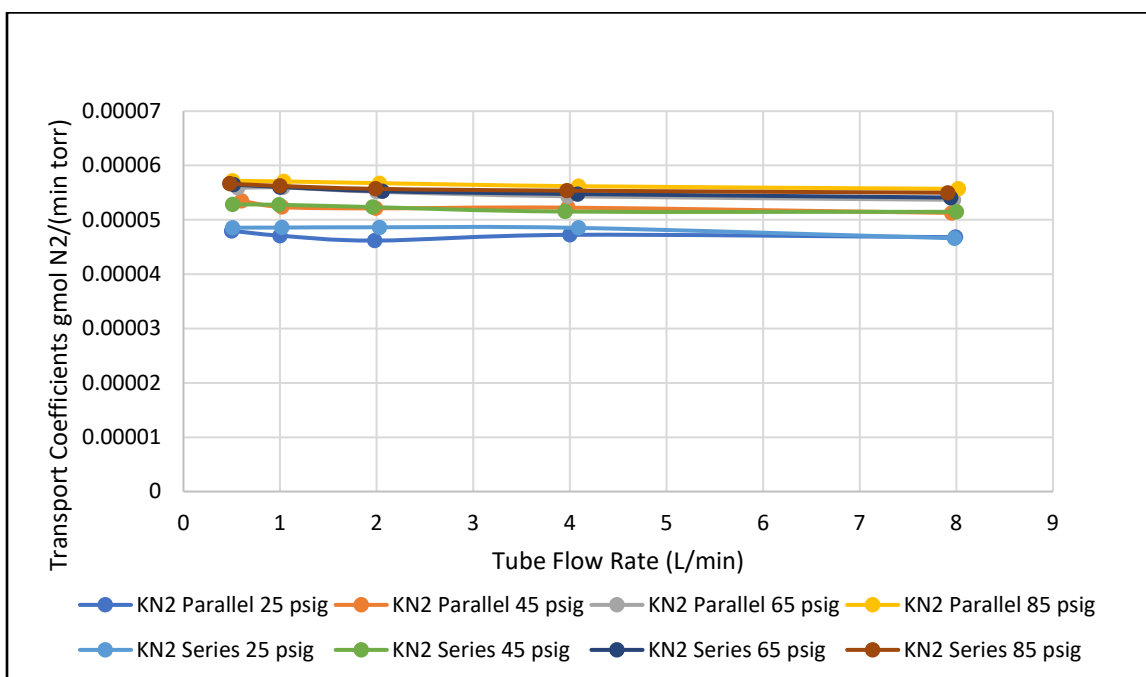


Figure 2: Nitrogen transport coefficients (parallel and series) VS tube flow rate

The Separation Factor was plotted based on the retentate and feed as a function of operating pressure 25psig, 45psig, 65psig, and 85psig with tube side flow rate as a parameter for the two configurations: parallel and series. The data is seen in **Appendix 1 Tables 3 through 6**. **Figure 3** below displays a pattern of higher separation factor with higher pressures. The series configuration of modules had higher separation factor when compared to the parallel for higher operating pressures, except for the lowest pressure in which they are relatively close. The lower operating pressures had a relatively constant separation factor when compared to higher ones like the 85psig. **Figure 4** below shows an increase of the separation factor for all operating pressures in both series and parallel by increasing the tube side flowrate of the feed. Therefore, by considering both **Figures 3 and 4**, the series configuration at higher operation operating pressure achieve the best separation.

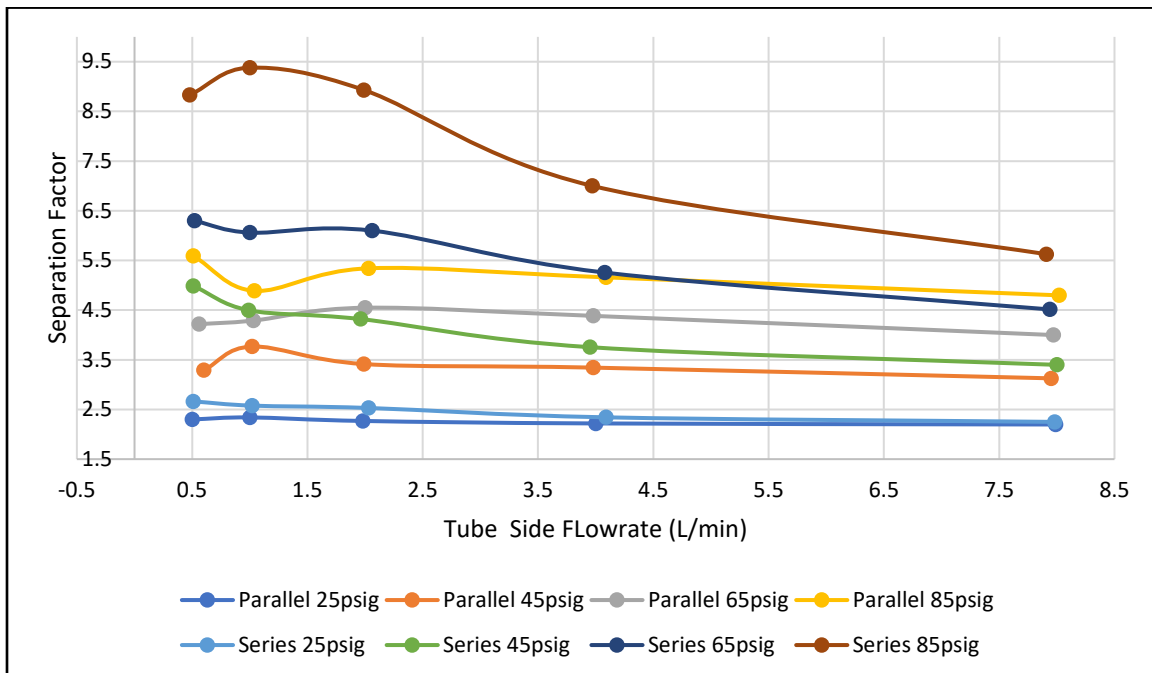


Figure 3: Separation factor (retentate-based) VS the tube flow rate

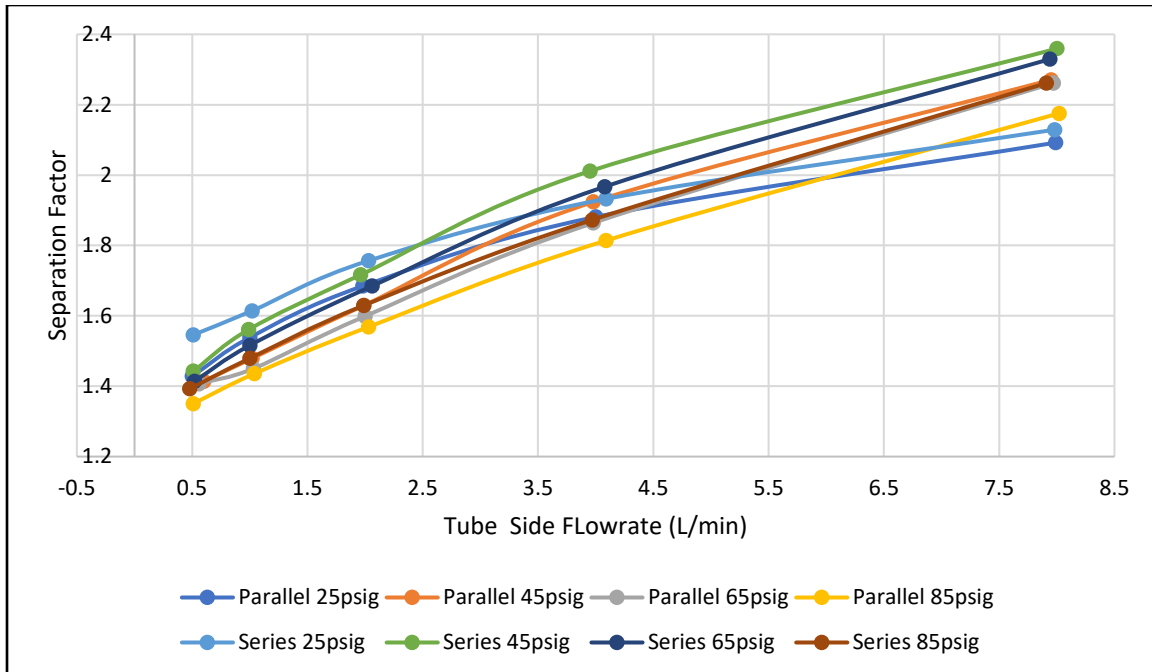


Figure 4: Separation factor (feed-based) VS tube flow rate

The Oxygen and Nitrogen recovery is plotted as a function of operating pressure with the tube flow rate as the parameter in **Figure 4** and **Figure 5** respectively. For the recovery of Oxygen, a higher yield is produced at higher pressure for both configuration modules of series and parallel at higher flow rates. For the recovery of Nitrogen, the yield is highest at lower pressure as the flow rate increases.

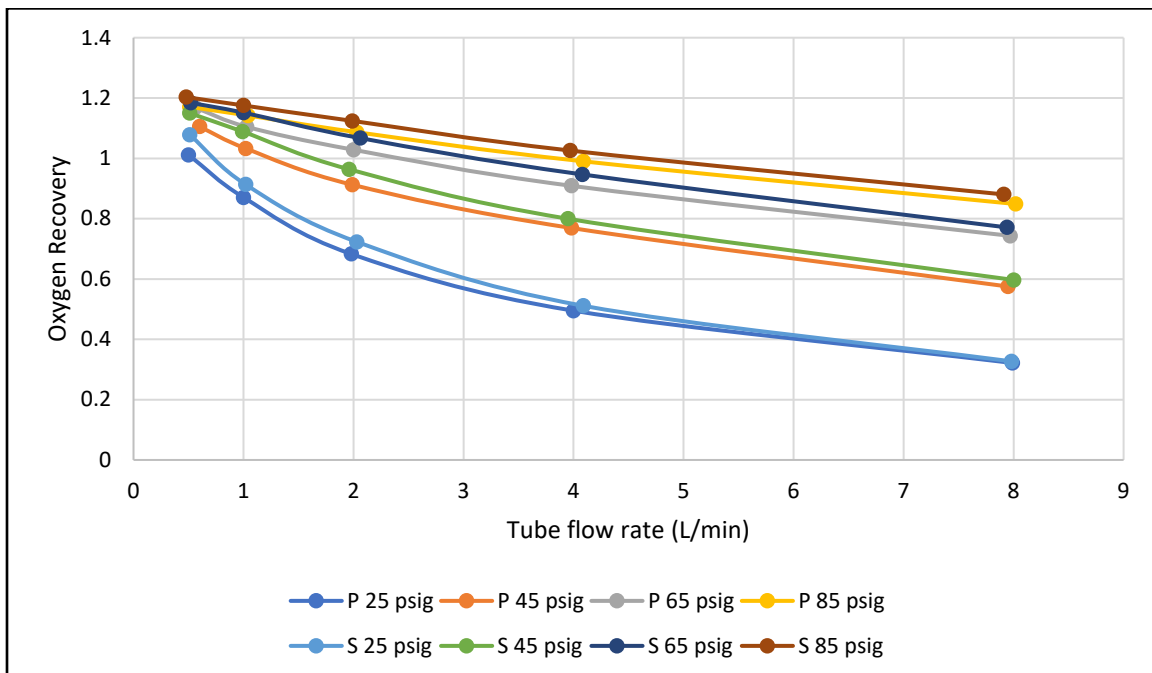


Figure 5: Oxygen recovery VS tube flow rate

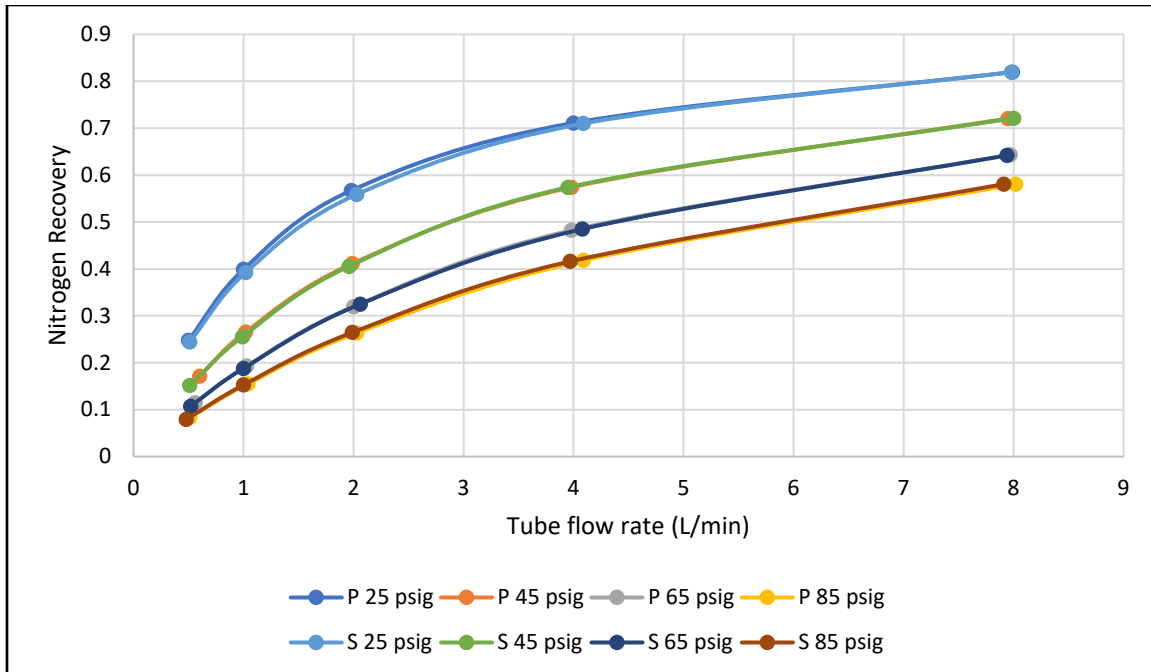


Figure 6: Nitrogen recovery VS tube flow rate

5. Discussion

From our results, it is shown that both oxygen and nitrogen transfer coefficients depend on the pressure and flow rate of air. Changes in pressure and flowrate affect the transfer coefficient in a logarithmic fashion, how they affect them is what differs. The transfer coefficient of nitrogen is more effected by the pressure of the system versus the flow rate, the opposite being true for Oxygen. One observation made is that by keeping the flow rate constant and varying the pressure of the system doesn't significantly change the transfer coefficient of oxygen. Conversely, maintaining the operating pressure of the system and varying the flow rate doesn't significantly affect the transfer coefficient of nitrogen. This relationship holds true in both series and parallel orientations.

As for the recovery of both nitrogen and oxygen, the results align with what one might expect when comparing the results to both Fick's and Henry's Law. Since the tube has an affinity for the oxygen, it makes more sense that an increase in the flow rate would propagate a lower recovery rate as the membrane gets saturated with oxygen. The oxygen's mean free path would decrease thus increasing the number of collisions, and subsequently the oxygen would take longer to escape the pipe. Essentially, Fick's law is at work. The more concentrated the tube is, the less oxygen is allowed to pass through. Another phenomenon we see is the pressure drop across the column for both oxygen and nitrogen are proportional to their mol fractions. Essentially, Henry's law is governing the results.

Progressing the experiment could mean inspecting how the pressure drop across the tube is a product of the systems pressure and the diameter. It's well studied that pressure drop is more prevalent at lower pressure with smaller diameters. How does the column participate in this phenomenon?

6. References

- [1] ChE 381: Unit Operations Laboratory. *Membrane Air Separation*. Fall 2018
- [2] Tuset, S. Condorchem Envitech. *Industrial Applications of Membrane Filtration*. Sep 2013.
- [3] Ismail, A.F.; Khulbe, K.C.; Matsuura, T. *Fundamentals of Gas Permeation Through Membranes*. Gas Separation Membranes. Pp. 11-35. DOI: https://doi.org/10.1007/978-3-319-01095-3_2

Appendices

7. Appendix I: Data Tabulation/Graphs

Table 1. Data Retrieved to Graph Transport Coefficients for Parallel Configurations

25 psig					
Parallel					
K O ₂	0.000137	0.000149	0.000158	0.000179	0.000197
Flow Rate (Tube)	0.5	1	1.98	4	7.99
K N ₂	4.8E-05	4.71E-05	4.62E-05	4.72E-05	4.68E-05
Flow Rate (Tube)	0.5	1	1.98	4	7.99

45 psig					
Parallel					
K O ₂	0.00015	0.000167	0.000173	0.000202	0.00022
Flow Rate (Tube)	0.6	1.02	1.99	3.98	7.95
K N ₂	5.35E-05	5.23E-05	5.21E-05	5.22E-05	5.13E-05
Flow Rate (Tube)	0.6	1.02	1.99	3.98	7.95

65 psig					
Parallel					
K O ₂	0.000158	0.000164	0.000183	0.000205	0.000228
Flow Rate (Tube)	0.56	1.03	2	3.98	7.97
K N ₂	5.59E-05	5.59E-05	5.52E-05	5.43E-05	5.37E-05
Flow Rate (Tube)	0.56	1.03	2	3.98	7.97

85 psig					
Parallel					
K O ₂	0.000161	0.000163	0.000183	0.000204	0.00023
Flow Rate (Tube)	0.51	1.04	2.03	4.09	8.02
K N ₂	5.72E-05	5.71E-05	5.68E-05	5.62E-05	5.57E-05
Flow Rate (Tube)	0.51	1.04	2.03	4.09	8.02

Table 2. Data Retrieved to Graph Transport Coefficients for Series Configurations

25 psig					
Series					
K O ₂	0.000176	0.000181	0.000197	0.000202	0.000206
Flow Rate (Tube)	0.51	1.02	2.03	4.09	7.98
K N ₂	4.86E-05	4.86E-05	4.86E-05	4.85E-05	4.66E-05
Flow Rate (Tube)	0.51	1.02	2.03	4.09	7.98

45 psig					
Series					
K O ₂	0.000191	0.000198	0.000214	0.000227	0.000245
Flow Rate (Tube)	0.51	0.99	1.96	3.95	8
K N ₂	5.28E-05	5.28E-05	5.24E-05	5.15E-05	5.15E-05
Flow Rate (Tube)	0.51	0.99	1.96	3.95	8

65 psig					
Series					
K O ₂	0.000188	0.000199	0.000221	0.00024	0.000255
Flow Rate (Tube)	0.52	1	2.06	4.08	7.94
K N ₂	5.65E-05	5.61E-05	5.53E-05	5.48E-05	5.41E-05
Flow Rate (Tube)	0.52	1	2.06	4.08	7.94

85 psig					
Series					
K O ₂	0.000189	0.000204	0.000222	0.000236	0.000255
Flow Rate (Tube)	0.48	1	1.99	3.97	7.91
K N ₂	5.67E-05	5.62E-05	5.57E-05	5.54E-05	5.5E-05
Flow Rate (Tube)	0.48	1	1.99	3.97	7.91

Table 3. Data Retrieved to Graph Separation Factor Based on Retentate and Feed Vs. Tube Side Flowrate at 25 psig

25psig												
Parallel							Series					
Mole fraction of O ₂ in tube	0.141	0.148	0.164	0.183	0.201		Mole fraction of O ₂ in tube	0.133	0.142	0.155	0.179	0.200
Mole fraction of O ₂ in shell	0.274	0.289	0.308	0.332	0.356		Mole fraction of O ₂ in shell	0.290	0.299	0.317	0.338	0.360
Mole fraction of N ₂ in tube	0.859	0.852	0.836	0.817	0.799		Mole fraction of N ₂ in tube	0.867	0.858	0.845	0.821	0.800
Mole fraction of N ₂ in shell	0.726	0.711	0.692	0.668	0.644		Mole fraction of N ₂ in shell	0.710	0.701	0.683	0.662	0.640
Separation Factor (Tube)	2.299	2.340	2.269	2.219	2.197		Separation Factor (Tube)	2.663	2.577	2.530	2.342	2.250
Separation Factor (Feed)	1.428	1.538	1.685	1.881	2.092		Separation Factor (Feed)	1.546	1.614	1.757	1.932	2.129

Table 4. Data Retrieved to Graph Separation Factor Based on Retentate and Feed Vs. Tube Side Flowrate at 45 psig

45psig												
Parallel							Series					
Mole fraction of O ₂ in tube	0.102	0.094	0.112	0.132	0.161		Mole fraction of O ₂ in tube	0.071	0.084	0.095	0.124	0.155
Mole fraction of O ₂ in shell	0.272	0.281	0.301	0.337	0.375		Mole fraction of O ₂ in shell	0.276	0.292	0.312	0.347	0.384
Mole fraction of N ₂ in tube	0.898	0.906	0.888	0.868	0.839		Mole fraction of N ₂ in tube	0.929	0.916	0.905	0.876	0.845
Mole fraction of N ₂ in shell	0.728	0.719	0.699	0.663	0.625		Mole fraction of N ₂ in shell	0.724	0.708	0.688	0.653	0.616
Separation Factor (Tube)	3.289	3.767	3.414	3.342	3.127		Separation Factor (Tube)	4.988	4.497	4.320	3.754	3.398
Separation Factor (Feed)	1.414	1.479	1.630	1.924	2.271		Separation Factor (Feed)	1.443	1.561	1.716	2.011	2.359

Table 5. Data Retrieved to Graph Separation Factor Based on Retentate and Feed Vs. Tube Side Flowrate at 65 psig

65psig												
Parallel							Series					
Mole fraction of O ₂ in tube	0.08 ₁	0.08 ₂	0.08 ₅	0.10 ₁	0.13 ₀		Mole fraction of O ₂ in tube	0.05 ₆	0.06 ₂	0.06 ₈	0.09 ₀	0.12 ₀
Mole fraction of O ₂ in shell	0.27 ₁	0.27 ₇	0.29 ₇	0.33 ₀	0.37 ₄		Mole fraction of O ₂ in shell	0.27 ₂	0.28 ₆	0.30 ₈	0.34 ₂	0.38 ₁
Mole fraction of N ₂ in tube	0.91 ₉	0.91 ₈	0.91 ₅	0.89 ₉	0.87 ₀		Mole fraction of N ₂ in tube	0.94 ₄	0.93 ₈	0.93 ₂	0.91 ₀	0.88 ₀
Mole fraction of N ₂ in shell	0.72 ₉	0.72 ₃	0.70 ₃	0.67 ₀	0.62 ₆		Mole fraction of N ₂ in shell	0.72 ₈	0.71 ₄	0.69 ₂	0.65 ₈	0.61 ₉
Seperation Factor (Tube)	4.21 ₈	4.28 ₉	4.54 ₈	4.38 ₄	3.99 ₈		Seperation Factor (Tube)	6.29 ₈	6.06 ₀	6.10 ₀	5.25 ₅	4.51 ₄
Seperation Factor (Feed)	1.40 ₇	1.45 ₀	1.59 ₉	1.86 ₄	2.26 ₁		Seperation Factor (Feed)	1.41 ₄	1.51 ₆	1.68 ₅	1.96 ₇	2.33 ₀

Table 6. Data Retrieved to Graph Separation Factor Based on Retentate and Feed Vs. Tube Side Flowrate at 85 psig

85psig												
Parallel							Series					
Mole fraction of O ₂ in tube	0.08 ₁	0.08 ₂	0.08 ₅	0.10 ₁	0.13 ₀		Mole fraction of O ₂ in tube	0.056	0.06 ₂	0.06 ₈	0.09 ₀	0.12 ₀
Mole fraction of O ₂ in shell	0.27 ₁	0.27 ₇	0.29 ₇	0.33 ₀	0.37 ₄		Mole fraction of O ₂ in shell	0.272	0.28 ₆	0.30 ₈	0.34 ₂	0.38 ₁
Mole fraction of N ₂ in tube	0.91 ₉	0.91 ₈	0.91 ₅	0.89 ₉	0.87 ₀		Mole fraction of N ₂ in tube	0.944	0.93 ₈	0.93 ₂	0.91 ₀	0.88 ₀

Mole fraction of N2 in shell	0.72 9	0.72 3	0.70 3	0.67 0	0.62 6	Mole fraction of N2 in shell	0.728	0.71 4	0.69 2	0.65 8	0.61 9
Separation Factor (Tube)	4.21 8	4.28 9	4.54 8	4.38 4	3.99 8	Separation Factor (Tube)	6.298	6.06 0	6.10 0	5.25 5	4.51 4
Separation Factor (Feed)	1.40 7	1.45 0	1.59 9	1.86 4	2.26 1	Separation Factor (Feed)	1.414	1.51 6	1.68 5	1.96 7	2.33 0

Table 7. Data Retrieved to graph The Oxygen and Nitrogen Recoveries in Parallel Configuration

25 psig					
Parallel					
O2	1.011689	0.870636	0.682927	0.494874	0.321648
Flow rate	0.5	1	1.98	4	7.99
N2	0.247938	0.398932	0.567112	0.711098	0.819371

45 psig					
Parallel					
O2	1.105239	1.032819	0.912386	0.76914	0.575083
Flow rate	0.6	1.02	1.99	3.98	7.95
N2	0.171147	0.265521	0.411424	0.573907	0.72072

65 psig					
Parallel					
O2	1.16836	1.104466	1.028497	0.908989	0.743098
Flow rate	0.56	1.03	2	3.98	7.97
N2	0.11495	0.193426	0.319548	0.48224	0.64314

85 psig					
Parallel					
O2	1.170339	1.142132	1.086755	0.990126	0.848577
Flow rate	0.51	1.04	2.03	4.09	8.02
N2	0.083137	0.154838	0.263742	0.417947	0.580396

Table 8. Data Retrieved to graph The Oxygen and Nitrogen Recoveries in Series Configuration

25 psig					
Series					
O2	1.07854	0.913163	0.723191	0.51113	0.32701
Flow Rate	0.51	1.02	2.03	4.09	7.98
N2	0.244105	0.392339	0.558913	0.709886	0.81937

45 psig					
Series					
O2	1.15007	1.089076	0.963721	0.79964	0.596935
Flow Rate	0.51	0.99	1.96	3.95	8
N2	0.151639	0.255334	0.405511	0.574076	0.721194

65 psig					
Series					
O2	1.184148	1.151556	1.067287	0.946657	0.771051
Flow Rate	0.52	1	2.06	4.08	7.94
N2	0.107553	0.18793	0.324927	0.484897	0.64196

85 psig					
Series					
O2	1.203027	1.175378	1.124207	1.025842	0.88037
Flow Rate	0.48	1	1.99	3.97	7.91
N2	0.079259	0.152661	0.264617	0.415946	0.580604

Table 9: Ratio of the oxygen coefficients to nitrogen coefficients

Pressure	Configuration	flowrates				
		~ 0.5	~ 1	~ 2	~ 4	~ 8
25 psig	parallel	2.854176	3.173941	3.423281	3.786347	4.203986
	series	3.634495	3.719318	4.058662	4.159538	4.41247
45 psig	parallel	2.807707	3.188232	3.326825	3.869412	4.280694
	series	3.607507	3.754787	4.093521	4.410117	4.749942
65 psig	parallel	2.819051	2.926509	3.313699	3.774697	4.249912
	series	3.32742	3.547252	3.998363	4.385211	4.711834
85 psig	parallel	2.821465	2.856038	3.21687	3.638696	4.128942
	series	3.330022	3.620282	3.98619	4.260978	4.634412

8. Appendix II: Error Analysis (10 pts)

Some errors have originated because we assumed that the air was 79.1% N₂ and 20.9% O₂ when air also contains other gases such as argon and carbon dioxide. Another source of error might be the temperature dependence of the permeability of the membrane in the system. In order to minimize this source of error, we conducted all the runs in a timely manner.

Another source of error was the oxygen sensor. The oxygen sensor did not calibrate to the oxygen content of air being 20.9%. This error may have caused some of the oxygen recovery values to be over 100%.

9. Appendix III: Sample Calculations (10 pts)

85psig Calculations

$$F(\text{tube side}) = \text{Flow Rate}(\text{tube}) * \frac{1000}{22400}$$

$$F(\text{tube side}) = 0.5 * \frac{1000}{22400} = .022321$$

$$G(\text{shell}) = \text{Flow Rate}(\text{shell}) * \frac{1000}{22400}$$

$$G(\text{shell}) = 0.075446 * \frac{1000}{22400} = 0.0034$$

Transport Coefficients:

$$KO_2 = \frac{O_2 \text{ flow rate}}{(O_2 \text{ tube pressure} - O_2 \text{ shell pressure})}$$

$$KO_2 = \frac{0.0207 \text{ gmol/min}}{(359.19 - 208.24)\text{torr}} = 0.000137$$

$$KN_2 = \frac{N_2 \text{ flow rate}}{(N_2 \text{ tube pressure} - N_2 \text{ shell pressure})}$$

$$KN_2 = \frac{0.0548 \text{ gmol/min}}{(1693.33 - 551.76)\text{torr}} = 0.000048$$

Separation Factor:

$$\alpha'_{AB} = \frac{\frac{X_{Ap}}{X_{Bp}}}{\frac{X_{Ar}}{X_{Br}}}$$

$$\alpha''_{AB} = \frac{\frac{X_{Ap}}{X_{Bp}}}{\frac{X_{Af}}{X_{Bf}}}$$

α'_{AB} = the separation factor based on retentate composition

α''_{AB} = the separation factor based on feed composition Recovery is defined by the Recovery equations below

X_{Ap} = the mole fraction of O_2 in permeate

X_{Bp} = the mole fraction of N_2 in permeate

X_{Ar} = the mole fraction of O_2 in retentate

X_{Br} = the mole fraction of N_2 in retentate

X_{Af} = the feed mole fraction of O_2

X_{Bf} = the feed mole fraction of N_2

$$\alpha'_{AB} = \frac{\frac{0.274}{0.726}}{\frac{0.141}{0.859}} = 2.299$$

$$\alpha''_{AB} = \frac{\frac{0.274}{0.726}}{\frac{0.209}{0.791}} = 1.428$$

Recoveries:

$$O_2 \text{ Recovery} = \left(\frac{0.274 * 1.69 \text{ L/min}}{0.209 * (0.5 + 1.69) \text{ L/min}} \right) = 1.012$$

$$N_2 \text{ Recovery} = \left(\frac{0.859 * 0.5 \text{ L/min}}{0.791 * (0.5 + 1.69) \text{ L/min}} \right) = 0.248$$

Ratio of Transport Coefficients:

$$Ratio = \frac{KO_2}{KN_2}$$

$$Ratio = \frac{0.000137}{0.000048} = 2.85$$

10. Appendix IV: Individual Team Contributions

NAME: Evelyn Almonte

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	In lab experiment
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)	1	
Materials & Supplies (prelab)	0.5	
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1.5	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	1.5	
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	2.5	
Error Analysis (final lab)		
Sample Calculations (final Lab)	2.5	
Power Point Presentation	0.5	
Total	13	

NAME: Sandra Hernandez

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	In lab experiment
Pre-Lab Editing	1.5	
Experimental Objective (Prelab)		
Apparatus (prelab)	0.5	
Materials & Supplies (prelab)	1	
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1.5	
Executive Summary (final lab)	1	
Introduction (Final lab)		
Theory (Final Lab)	1	
Results (final lab)	1	
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1.5	
Error Analysis (final lab)		
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	13	

NAME: Tianna Mitchell

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	In lab experiment
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1.5	
Executive Summary (final lab)		
Introduction (Final lab)	0.5	
Theory (Final Lab)	0.5	
Results (final lab)	1	
Discussion (final lab)		
References (final and/or prelab)	0.5	
Data Tabulation/Graphs (final)	2	
Error Analysis (final lab)		
Sample Calculations (final Lab)	2.5	
Power Point Presentation	0.5	
Total	13	

NAME: Anne Serban

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	In lab experiment
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	1	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1.5	
Executive Summary (final lab)	0.5	
Introduction (Final lab)		
Theory (Final Lab)	0.5	
Results (final lab)	1	
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)	0.5	
Sample Calculations (final Lab)	2.5	
Power Point Presentation	0.5	
Total	13	

NAME: Maurice Smith

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	In lab experiment
Pre-Lab Editing	1	
Experimental Objective (Prelab)	0.5	
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	1	
Project Safety Evaluation (prelab)		
Final Lab Editing	1.5	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	1	
Discussion (final lab)	2	
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)		
Sample Calculations (final Lab)	2.5	
Power Point Presentation	0.5	
Total	13	